AIR FRYER MARKET ANALYSIS

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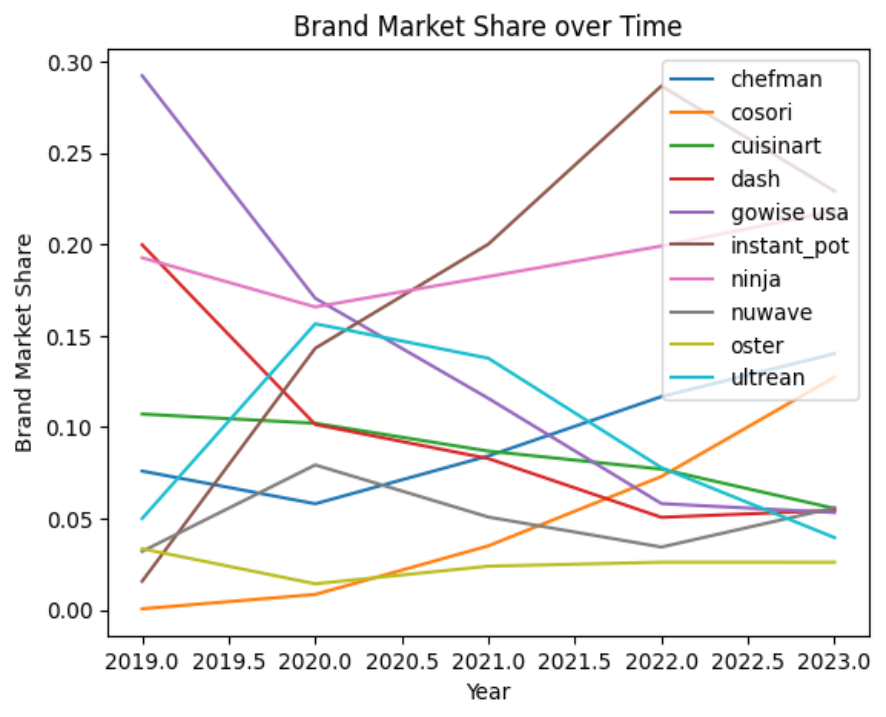
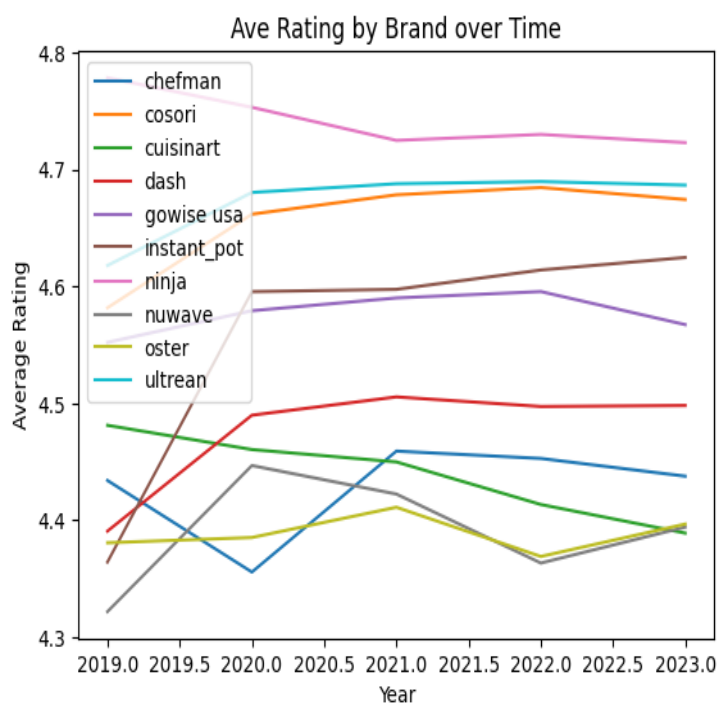
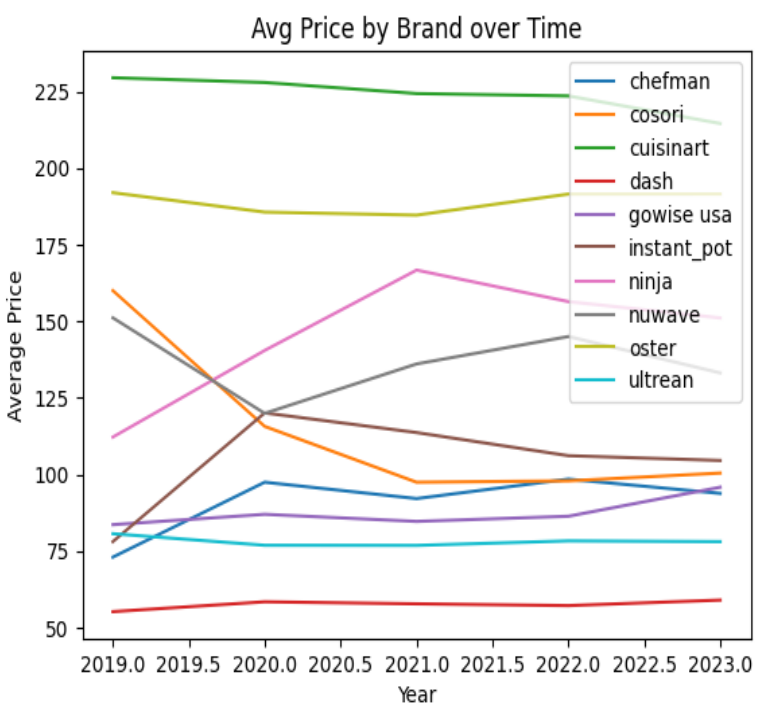
MARKET OVERVIEW + TRENDS

- Market includes premium (Cuisinart, Oster) and mass brands (Ninja, Instant Pot)
- Ninja dominates with strong ratings + share
- Instant Pot grew rapidly by lowering price and improving ratings
- Ratings are consistently high -> price does not equal quality necessarily







DEMAND MODEL

- We estimate demand using a multinomial logit model:

$$\log(s_{bt}) = \alpha + \beta_{price}\, p_{bt} + \beta_{rating}\, r_{bt} + \text{features} + \text{brand effects} + \text{year}$$

- We model how market share depends on price, ratings, and product features
- Brand and year effects control for:
  - Brand reputation
  - Market trends over time

PRICE EFFECTS + FEATURES

Estimated price coefficient:

$$\beta_price = -0.0377$$

Interpretation:

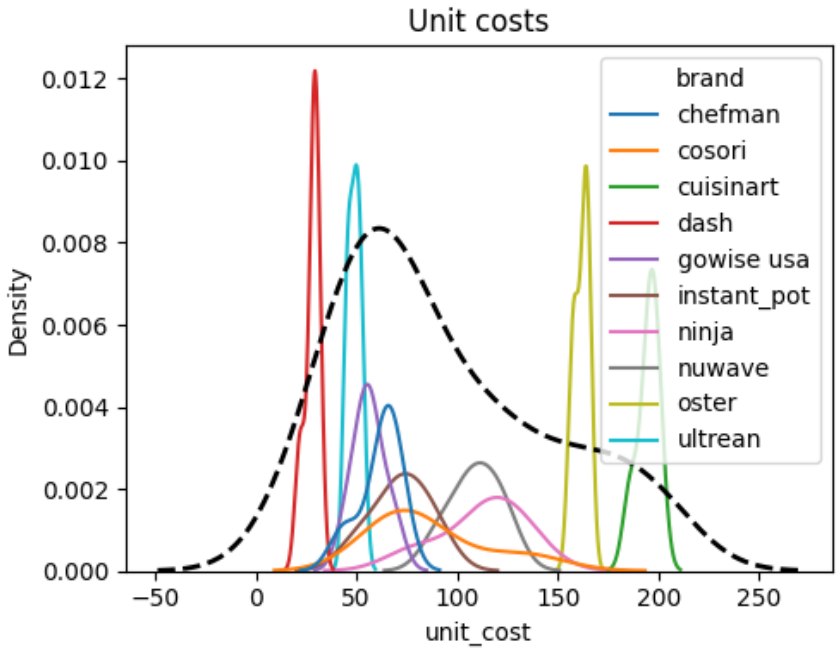
1. Higher prices ➡ Lower Demand (expected)
2. Market is price-sensitive

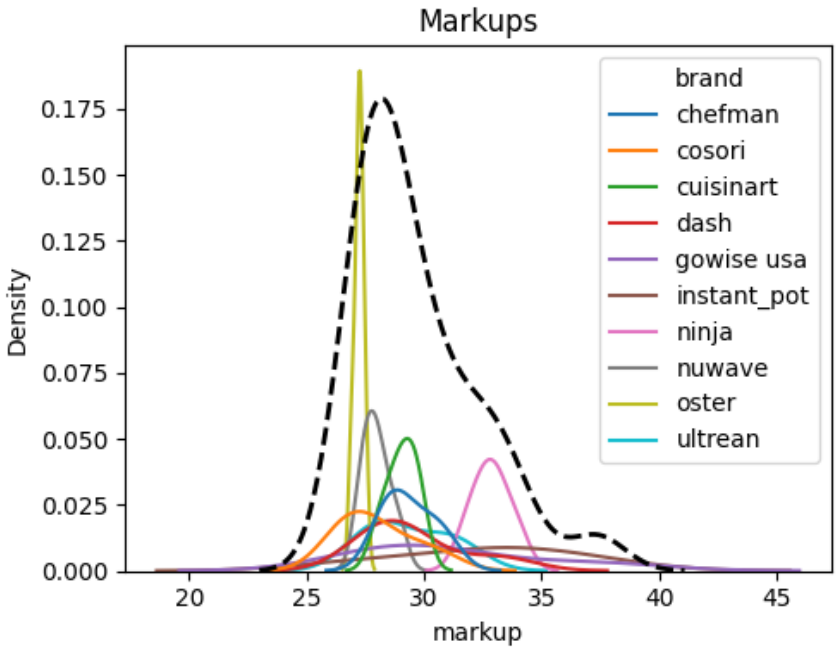
Top Demand Drivers:

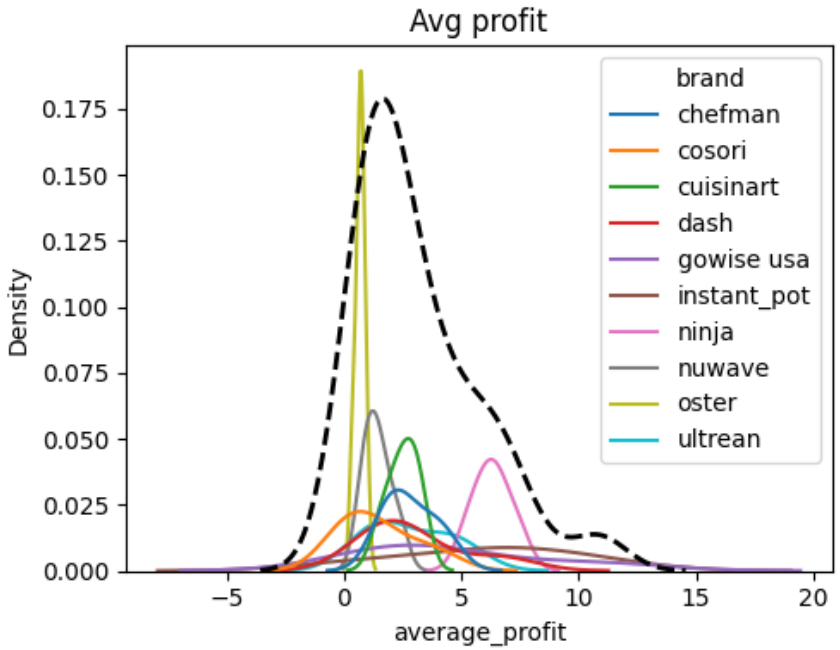
1. Compact designs
2. Oven-style models
3. Viewing windows

COSTS, MARKUPS, PROFITS

- Premium brands show higher markups but lower market share
- Mass-market brands (e.g., Ninja) achieve higher share-weighted profits
- Profit is measured as share-weighted profit derived from the demand model
- Cost and price differences reflect product positioning strategy







CONCLUSION

- Market is competitive + price-sensitive
- Mid-priced brands dominate demand, balancing quality and affordability
- Winning strategy:
  - Competitive pricing
  - Strong consumer ratings
  - Practical, high-value features
- Ninja = Strongest Overall Position
- Instant Pot = Fastest Growth Trajectory